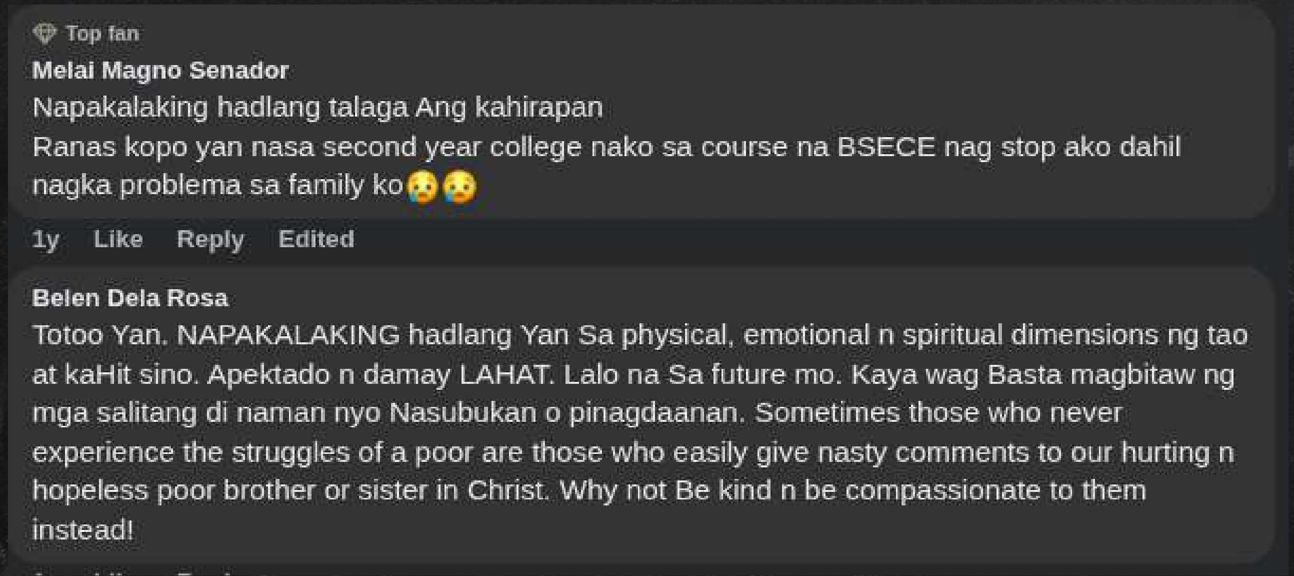
A Regional Correlation Study of Household Economics and School Enrollment in the Philippines

Presented By Group 20 : Hans Jio Arca

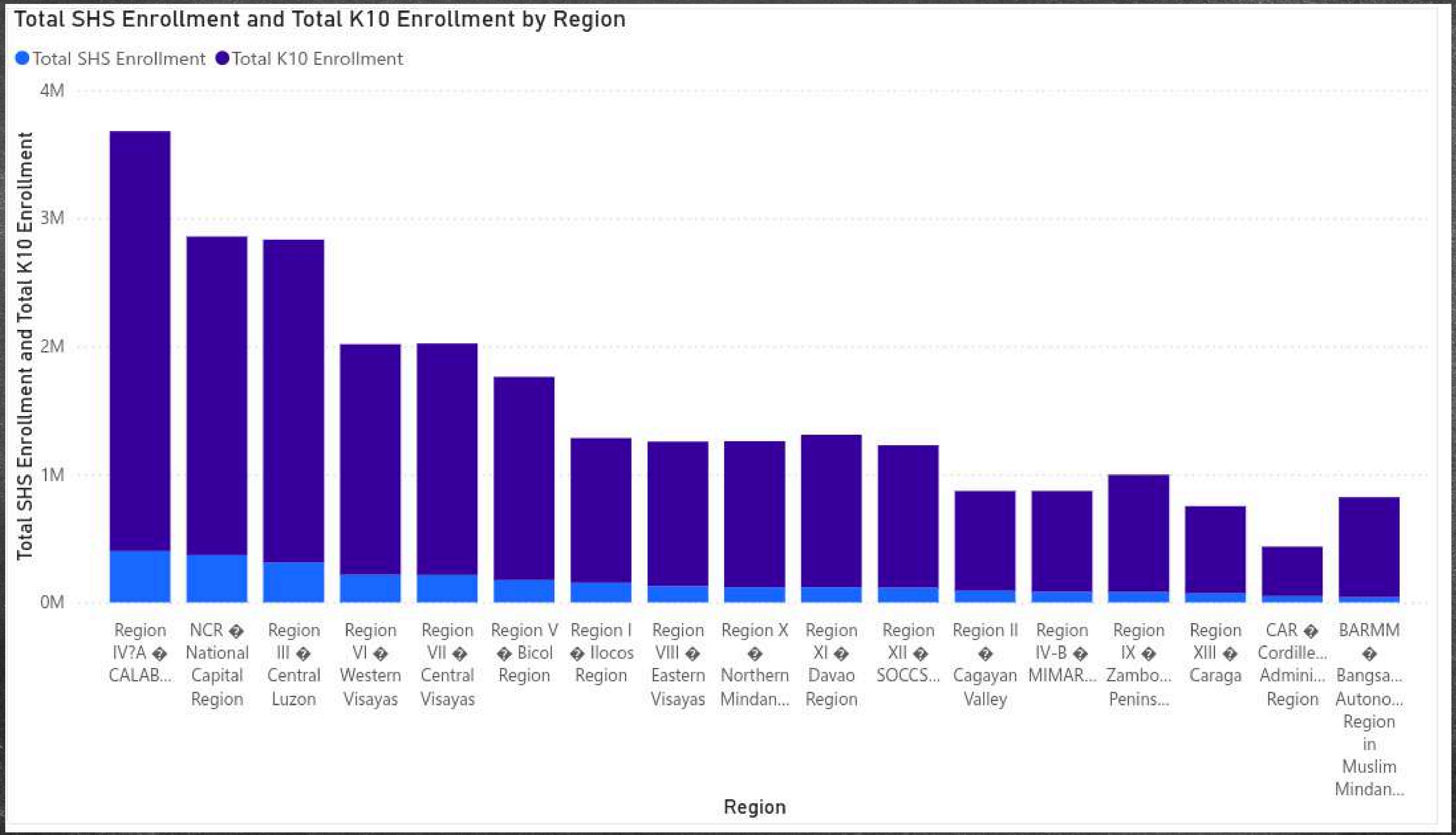
Introduction of Research Gap



While the K-12 program was designed to universally expand educational access, there is a critical lack of quantitative research explaining the significant drop in enrollment volume as students transition from K-10 to Senior High School. Current discourse often relies on the narrative of 'resilience,' yet few studies have statistically analyzed how household economic survival (specifically high Food Share) acts as a filtering mechanism, directly correlating with the dwindling number of students who decide to pursue and remain in specialized academic strands versus those who are forced to stop or shift to vocational tracks.

Project Objectives & Findings

Objective 1. How are K–10 and SHS enrollments distributed across regions, based on the available aggregated enrollment data?



Project Objectives & Findings

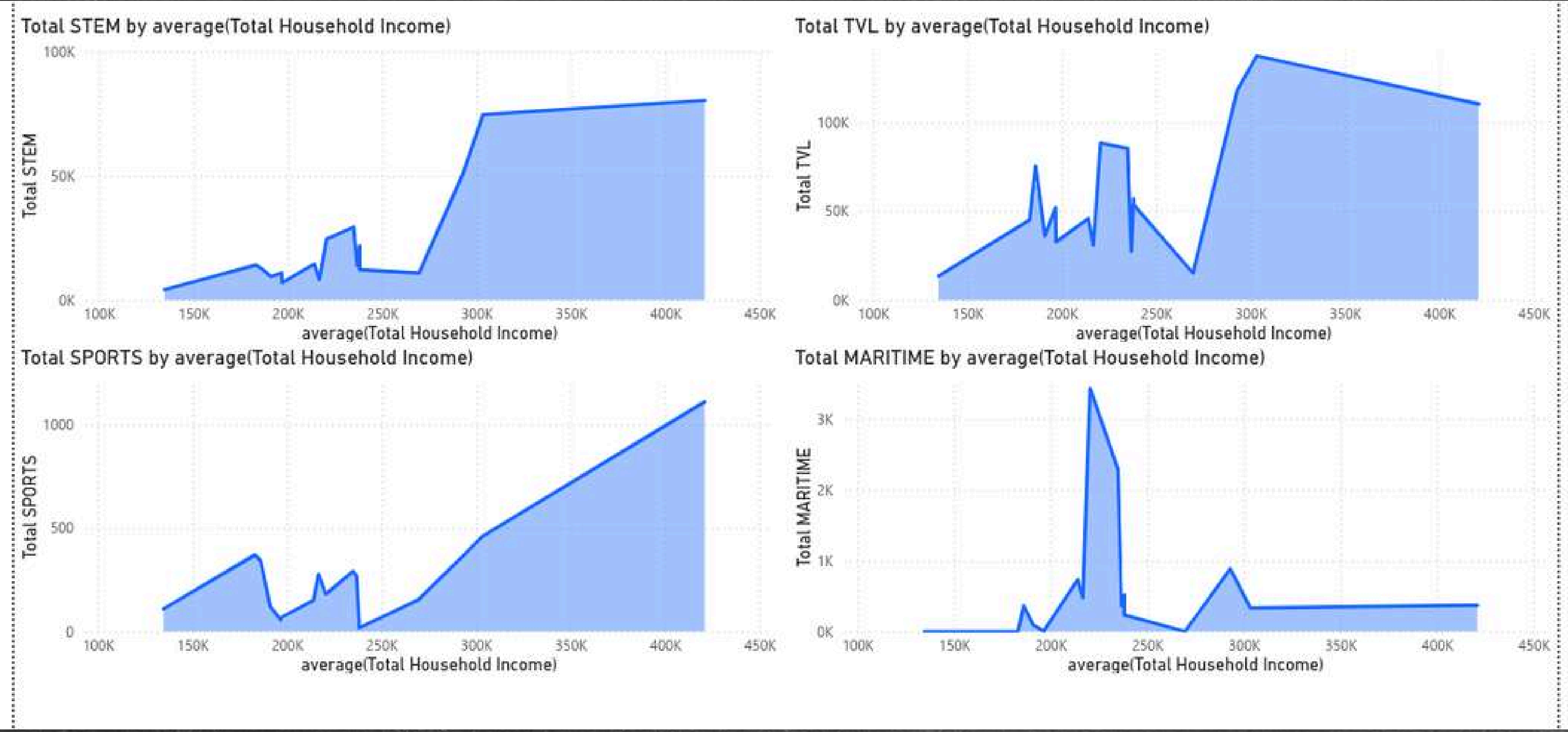
Question 2 & 3. Which socioeconomic factors most strongly influence SHS strand selection? What correlations exist between household socioeconomic variables and school enrollment indicators, such as K–10 totals, SHS totals, and SHS strand counts?

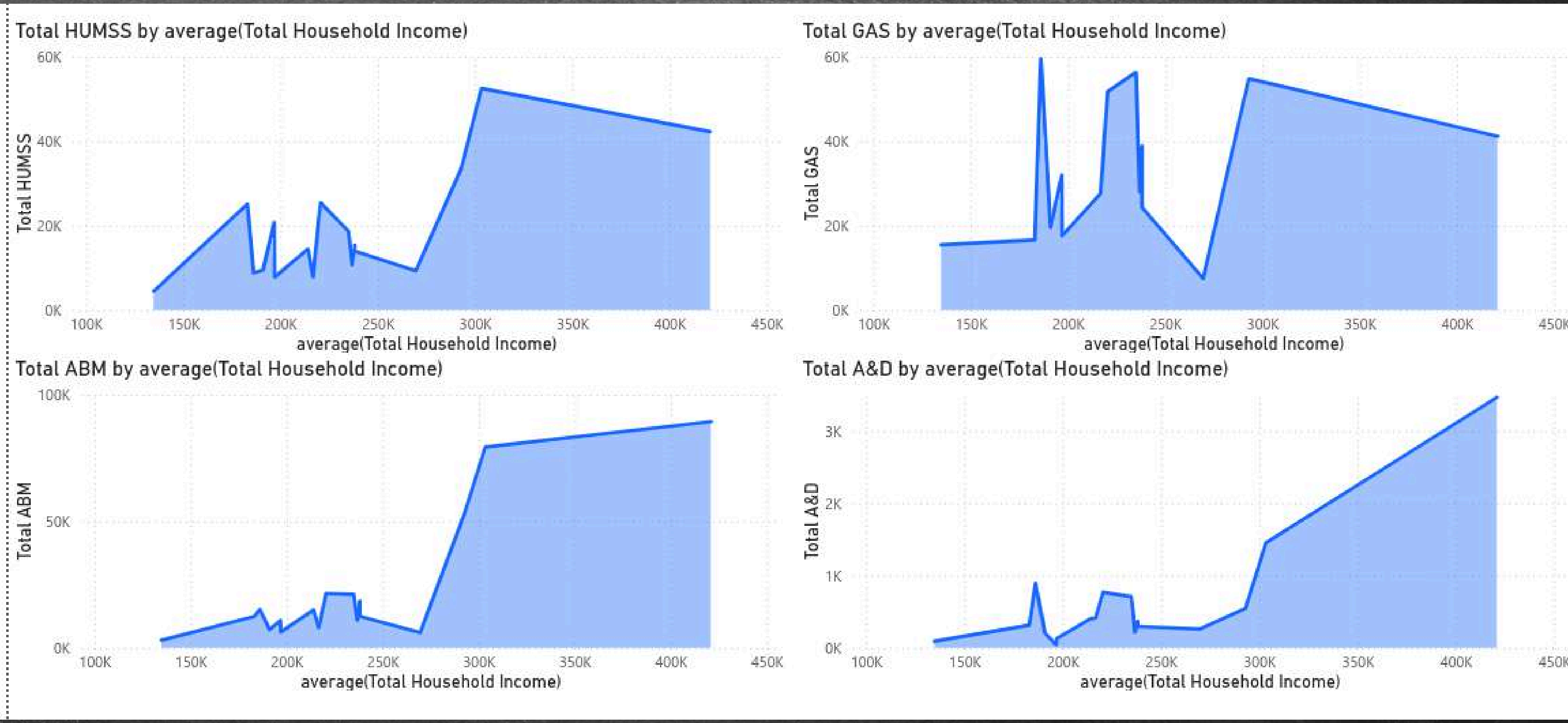
Correlation Matrix (Correlation Matrix (2))

Attributes	average...	average...	average...	average...	average...	average...	K10_To...	Total_S...	ABM_T...	HUMSS...	STEM_...	GAS_T...	TVL_To...	MARITI...	SPORT...	A&D_T...
average(Total Household Income)	1	0.891	-0.712	0.712	0.935	-0.488	0.245	0.433	0.540	0.463	0.506	0.131	0.338	0.017	0.542	0.553
average(Education Expenditure)	0.891	1	-0.622	0.622	0.810	-0.370	0.205	0.361	0.464	0.399	0.431	0.080	0.280	-0.036	0.463	0.473
average(Food Share)	-0.712	-0.622	1	-1.000	-0.460	0.670	-0.053	-0.151	-0.267	-0.219	-0.245	0.067	-0.077	0.102	-0.252	-0.256
average(Non-Food Share)	0.712	0.622	-1.000	1	0.460	-0.670	0.053	0.151	0.267	0.219	0.245	-0.067	0.077	-0.102	0.252	0.256
average(Total Food Expenditure)	0.935	0.810	-0.460	0.460	1	-0.315	0.310	0.506	0.592	0.513	0.553	0.209	0.420	0.043	0.563	0.581
average(Total Number of Family members)	-0.488	-0.370	0.670	-0.670	-0.315	1	-0.081	-0.170	-0.165	-0.198	-0.162	-0.082	-0.159	-0.079	-0.112	-0.129
K10_Total	0.245	0.205	-0.053	0.053	0.310	-0.081	1	0.585	0.235	0.516	0.118	0.776	0.791	-0.128	0.603	0.296
Total_SHS	0.433	0.361	-0.151	0.151	0.506	-0.170	0.585	1	0.887	0.959	0.839	0.818	0.941	0.172	0.761	0.812
ABM_Total	0.540	0.464	-0.267	0.267	0.592	-0.165	0.235	0.887	1	0.898	0.982	0.476	0.691	0.220	0.674	0.871
HUMSS_Total	0.463	0.399	-0.219	0.219	0.513	-0.198	0.516	0.959	0.898	1	0.847	0.695	0.877	0.142	0.681	0.759
STEM_Total	0.506	0.431	-0.245	0.245	0.553	-0.162	0.118	0.839	0.982	0.847	1	0.412	0.610	0.320	0.584	0.832
GAS_Total	0.131	0.080	0.067	-0.067	0.209	-0.082	0.776	0.818	0.476	0.695	0.412	1	0.927	0.060	0.664	0.527
TVL_Total	0.338	0.280	-0.077	0.077	0.420	-0.159	0.791	0.941	0.691	0.877	0.610	0.927	1	0.049	0.741	0.662
MARITIME_Total	0.017	-0.036	0.102	-0.102	0.043	-0.079	-0.128	0.172	0.220	0.142	0.320	0.060	0.049	1	-0.091	0.177
SPORTS_Total	0.542	0.463	-0.252	0.252	0.563	-0.112	0.603	0.761	0.674	0.681	0.584	0.664	0.741	-0.091	1	0.838
A&D_Total	0.553	0.473	-0.256	0.256	0.581	-0.129	0.296	0.812	0.871	0.759	0.832	0.527	0.662	0.177	0.838	1

Project Objectives & Findings

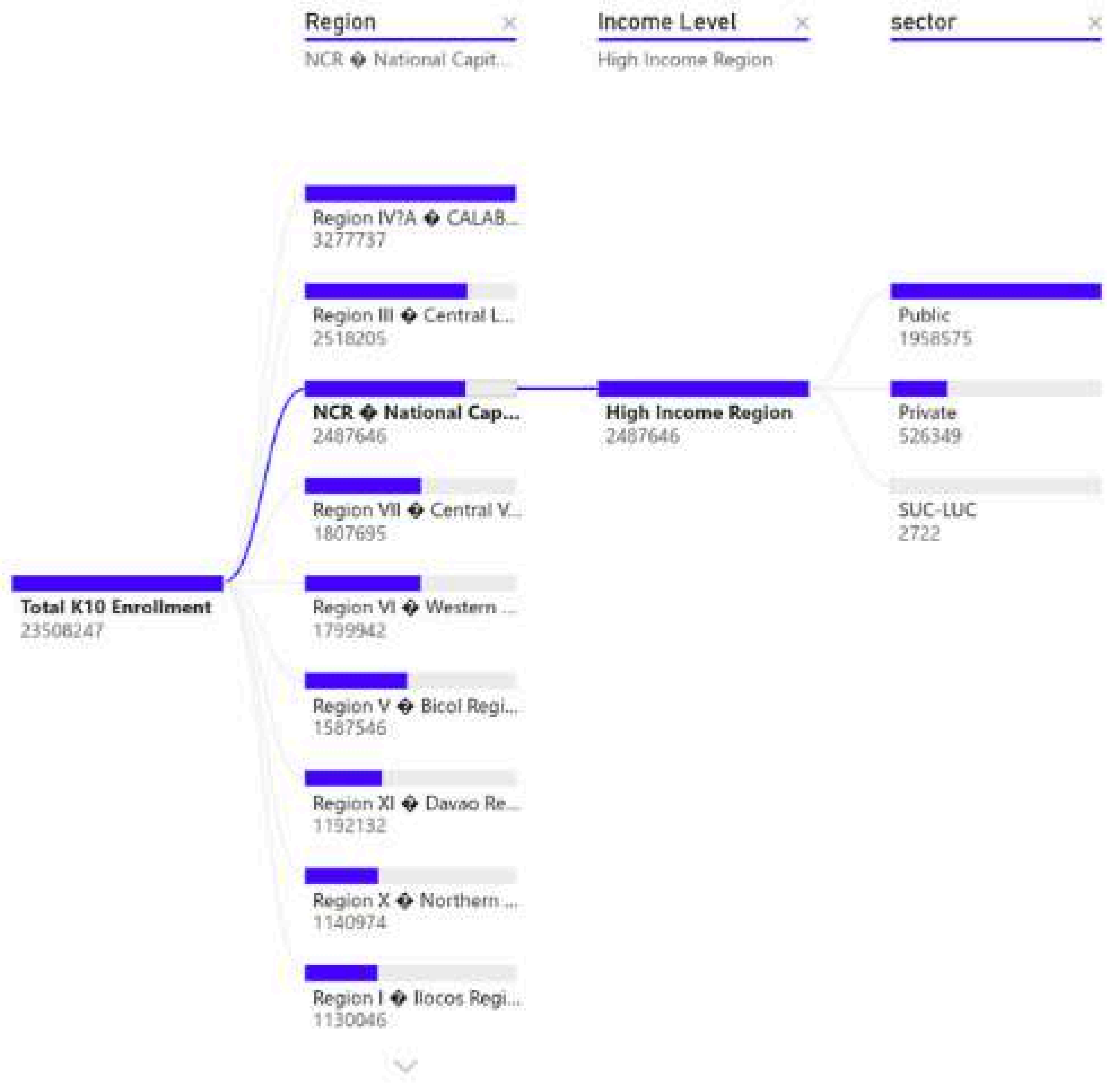
Question 4. Which SHS strands do students from regions with higher household income or higher education expenditure more likely to pursue?



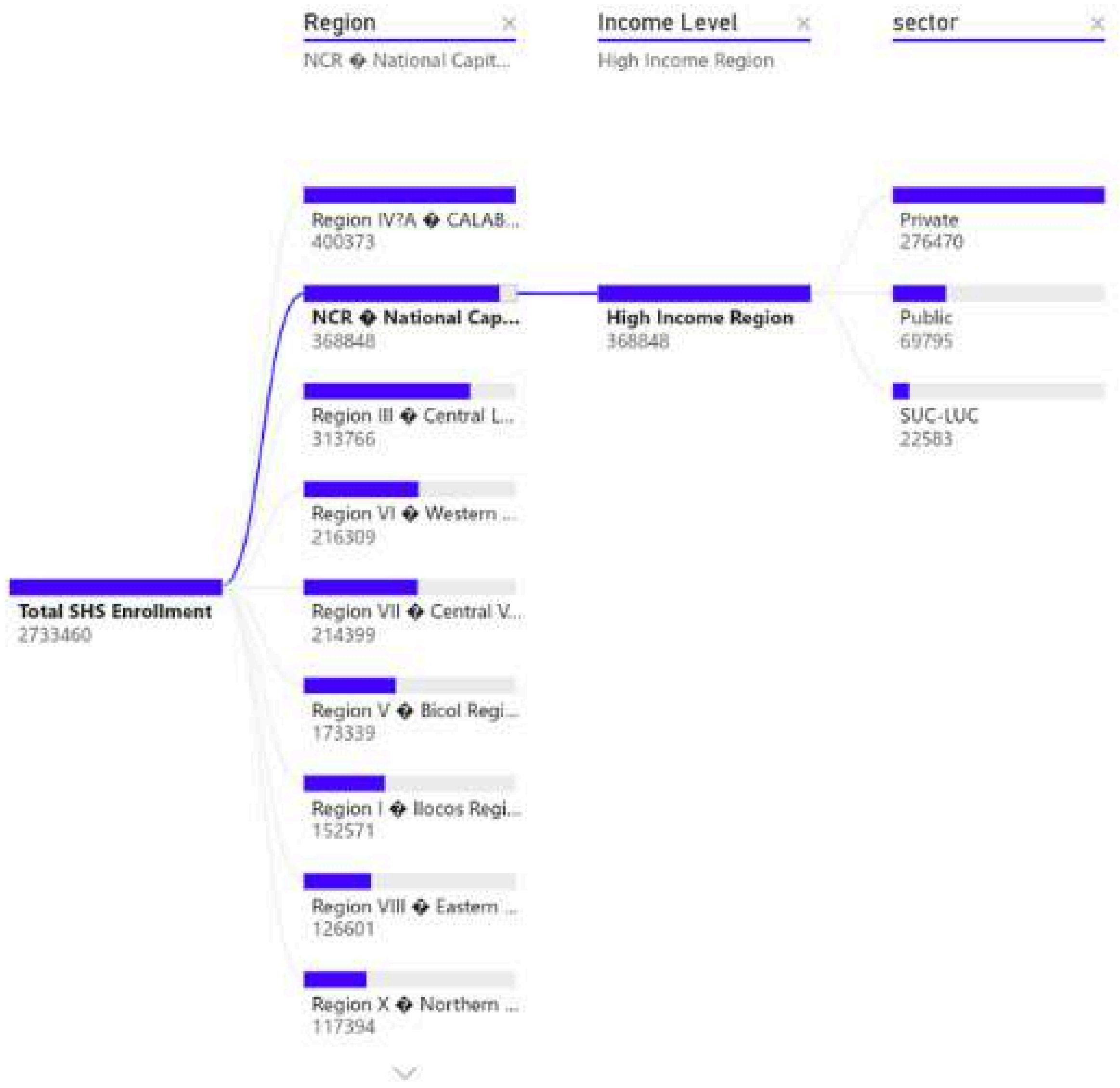


Project Objectives & Findings

Question 5. & 6. Are there observable associations between income levels and the type of academic sector students enroll in? What insights can be derived from the combined economic–education data that may help policymakers understand regional disparities in schooling and resource needs?



Decomposition of K10 Enrollment Across Regions



Decomposition of SHS Enrollment Across Regions

Main Dashboard

A REGIONAL CORRELATION STUDY OF HOUSEHOLD ECONOMICS AND SCHOOL ENROLLMENT IN THE PHILIPPINES IN 2017

DATASET OVERVIEW & PROFILE |Descriptive Analytics

Average Household Income (Annual)

233.79K

Average of average(Total Household Income)

Average Household Expenditure (Annual)

159.87K

Total Avg Expenditure

Total Enrollees

26.242M

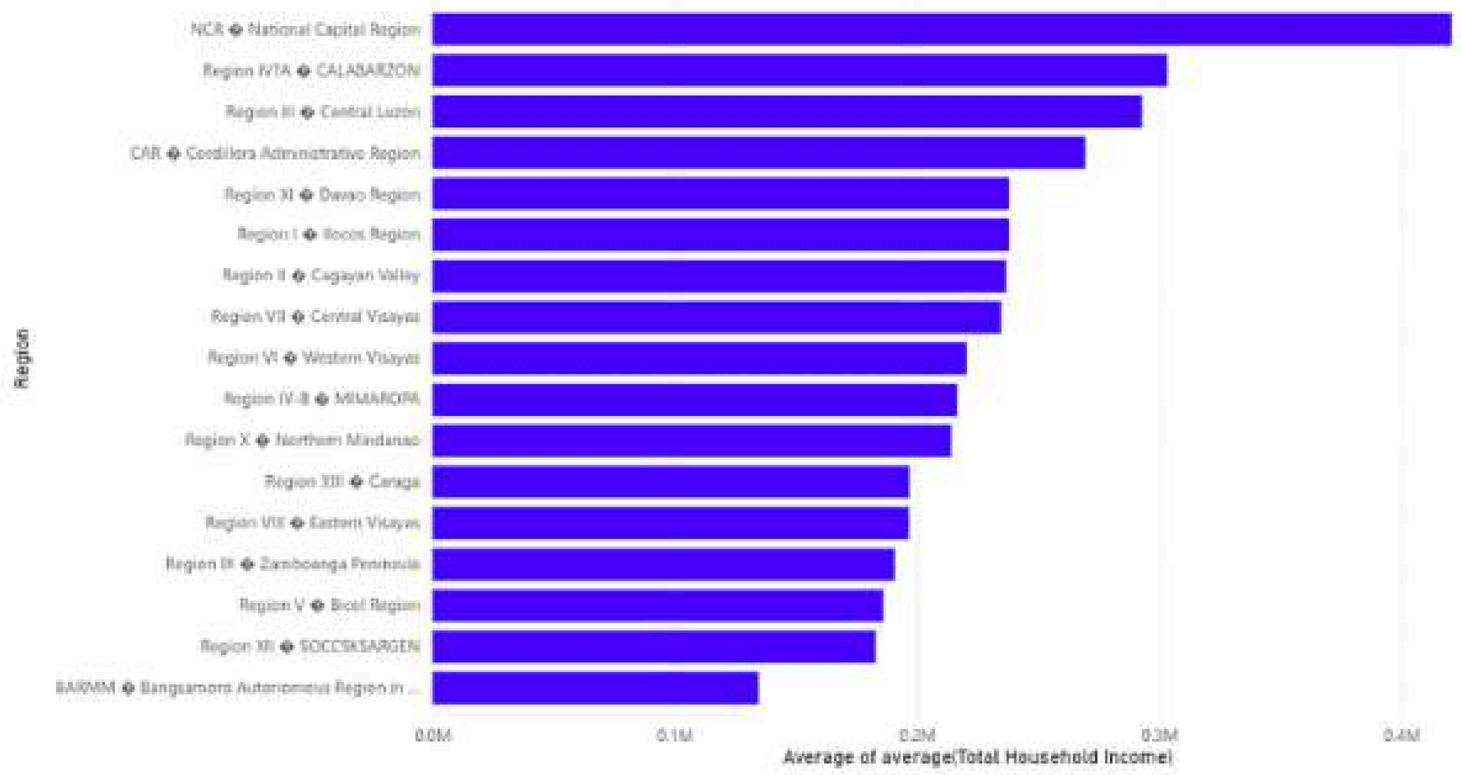
Total Enrollees

Average Expenditure in Education (%)

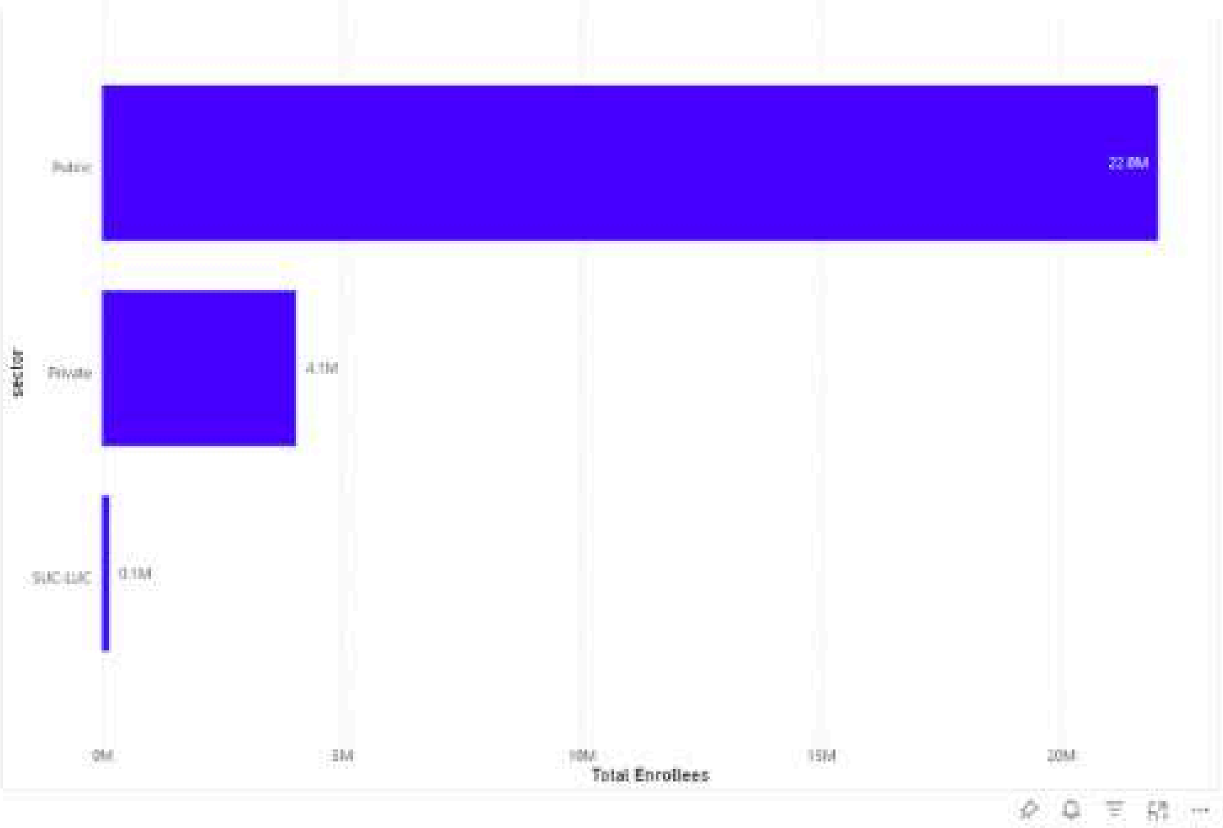
0.04

Education Share

Distribution of Total Household Income Grouped by Region



Number of Enrollees per Educational Sector



Recommendations

Establish Needs-Based Subsidies for Academic Strands

- Create targeted financial support programs for students in economically vulnerable regions.
- Ensure that enrollment in resource-intensive tracks (like STEM and ABM) is driven by student aptitude rather than household financial capacity.

Modernize and Standardize Public TVL Education

- Upgrade facilities and integrate industry-recognized certifications into the public school TVL curriculum.
- Transform the technical-vocational pathway into a high-value, employable option rather than just a "low-cost" alternative.

Adopt Socioeconomic-Based Resource Allocation

- Shift from uniform funding to a model that prioritizes resources based on regional economic profiles.
- Direct funding specifically toward retention programs for struggling regions and academic expansion for specialized hubs.

Thank you